

JavaScript Operators

| What is an Operator?

An **operator** is a symbol that performs an operation on one or more values (operands).

Example

```
let sum = 10 + 5;  
console.log(sum); // 15
```

Here:

- `10` and `5` are **operands**.
- `+` is the **operator**.

Types of Operators

1. Arithmetic Operators
 2. Assignment Operators
 3. Comparison Operators
 4. Logical Operators
 5. Increment & Decrement Operators
 6. String Operators
 7. Ternary Operator
 8. Type Operators
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1. Arithmetic Operators

Used to perform mathematical calculations.

Operator	Meaning	Example	Result
+	Addition	10 + 5	15
-	Subtraction	10 - 5	5
*	Multiplication	10 * 5	50
/	Division	10 / 5	2
%	Modulus (Remainder)	10 % 3	1
**	Exponent	2 ** 3	8

EXAMPLE

```
let a = 10;
let b = 3;

console.log(a + b);
console.log(a - b);
console.log(a * b);
console.log(a / b);
console.log(a % b);
console.log(a ** b);
```

2. Assignment Operators

Used to assign values.

Operator	Example	Same As
=	x = 5	Assign value
+=	x += 2	x = x + 2
-=	x -= 2	x = x - 2
*=	x *= 2	x = x * 2
/=	x /= 2	x = x / 2
%=	x %= 2	x = x % 2

EXAMPLE

```
let x = 10;

x += 5;
console.log(x); // 15

x -= 3;
console.log(x); // 12
```

3. Comparison Operators

Used to compare two values.

Operator	Meaning	Example
==	Equal (value only)	5 == "5" → true
===	Strict Equal	5 === "5" → false
!=	Not Equal	5 != 4
!==	Strict Not Equal	5 !== "5"
>	Greater Than	10 > 5
<	Less Than	5 < 10
>=	Greater Than or Equal	5 >= 5
<=	Less Than or Equal	5 <= 10

EXAMPLE

```
console.log(10 > 5);
console.log(10 === "10");
console.log(10 == "10");
```

4. Logical Operators

Used to combine multiple conditions.

Operator	Meaning
&&	AND
	OR
!	NOT

EXAMPLE

```
let age = 20;
let hasID = true;

console.log(age >= 18 && hasID);
console.log(age < 18 || hasID);
console.log(!hasID);
```

5. Increment & Decrement Operators

Increase or decrease a value by 1.

Operator	Meaning
++	Increment
--	Decrement

EXAMPLE

```
let count = 5;

count++;
console.log(count); // 6

count--;
console.log(count); // 5
```

6. String Operator

The  operator joins strings.

EXAMPLE

```
let firstName = "MD";  
let lastName = "Rokonuzzaman";  
  
console.log(firstName + " " + lastName);
```

Output

```
MD Rokonuzzaman
```

7. Ternary Operator

A short way to write an if...else statement.

SYNTAX

```
condition ? trueValue : falseValue;
```

EXAMPLE

```
let age = 20;

let result = age >= 18 ? "Adult" : "Minor";

console.log(result);
```

8. Type Operator

The `typeof` operator returns the data type.

EXAMPLE

```
console.log(typeof "Hello");
console.log(typeof 100);
console.log(typeof true);
console.log(typeof []);
console.log(typeof {});
```

Operator Priority (Precedence)

```
let result = 10 + 5 * 2;
console.log(result);
```

Output

```
20
```

Because multiplication () has higher precedence than addition ().

Summary

Category	Purpose
Arithmetic	Mathematical calculations
Assignment	Assign values
Comparison	Compare values
Logical	Combine conditions
Increment/Decrement	Increase or decrease by 1
String	Join strings
Ternary	Short if...else
Type	Check data type

Easy Way to Remember

-     → **Arithmetic**
- = += -= → **Assignment**
- == === > < → **Comparison**
- && || ! → **Logical**
- ++ -- → **Increment / Decrement**

- ◦ (String) → **Concatenation**
 - `?:` → **Ternary**
 - `typeof` → **Type Checking**
-

Interview Questions

Q1. WHAT IS THE DIFFERENCE BETWEEN `==` AND `===`?

- `==` compares only values (type conversion happens).
- `===` compares both value and data type.

```
console.log(5 == "5"); // true
console.log(5 === "5"); // false
```

Q2. WHAT IS THE DIFFERENCE BETWEEN `&&` AND `||`?

- `&&` (AND): All conditions must be true.
- `||` (OR): At least one condition must be true.

```
console.log(true && false); // false
console.log(true || false); // true
```